



Tribhuvan University
Faculty of Humanities and Social Science

A Project Report on
Online Bus Reservation System

Submitted to
Department of Computer Application
Nepalaya College
Kalanki, Kathmandu

*In Partial fulfilment of the requirements of Bachelors in Computer
Application*

Submitted By
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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project proposal under my supervision by **Udesh Regmi & Sangam Sedai** entitled “**Online Bus Reservation System**” in partial fulfilment of the requirements for the degree of Bachelor in Computer Application is recommended for the final evaluation.

SIGNATURE

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LETTER OF APPROVAL

This is to certify that this project proposed by **“Udesh Regmi and Sangam Sedai”** entitled **“Online Bus Reservation System”** in partial fulfilment of the requirements for the degree Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

The rapid advancement in information technology has revolutionized the transportation sector, particularly in making travel more accessible and efficient. This project presents the development of an Online Bus Reservation System (OBRS), a web-based application designed to streamline the process of booking bus tickets for passengers while providing administrative tools for bus operators. Traditional bus reservation methods often involve long queues, manual record-keeping, and limited availability, leading to inefficiencies and customer dissatisfaction. OBRS addresses these challenges by offering a user-friendly online platform where passengers can search for available buses, view routes and schedules, select seats, make secure payments, and receive e-tickets instantly. The system is built using PHP as the backend scripting language, MySQL for database management, and HTML/CSS/JavaScript for the frontend interface, ensuring responsiveness across devices. For administrators, the system includes modules for managing bus details, user accounts, bookings, and generating reports. The implementation follows a structured methodology, incorporating feasibility studies, entity-relationship modeling, and data flow diagrams to ensure robustness. Through this system, the project aims to reduce booking time by 70%, enhance data accuracy, and improve overall user satisfaction. Extensive testing validates the system's reliability, security, and scalability, making it suitable for small to medium-scale bus service providers. This work contributes to the digital transformation of public transport, promoting paperless operations and real-time updates.

Keywords: Online Bus Reservation, Web-based Application, PHP MySQL, Seat Booking, Admin Panel, E-Ticket Generation, Transportation Management

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project supervisor **Mr. Yuba Raj Devkota** for his continuous guidance, valuable suggestions, and constant encouragement throughout the completion of this project. His expertise, insightful feedback, and timely support played a crucial role in shaping the project and ensuring its successful completion.

We are also thankful to the Head of the Department and all the faculty members of Nepalaya College for providing the necessary academic support, facilities, and learning environment required for carrying out this project effectively. Their cooperation and guidance greatly contributed to enhancing our understanding of the subject matter.

We would like to extend our heartfelt thanks to our friends and family members for their moral support, patience, and encouragement during the entire project duration. Their motivation helped us remain focused and committed to our work.

Finally, we express our gratitude to Tribhuvan University for including *Project-I* in the BCA 4th Semester curriculum, which provided us with an opportunity to gain practical exposure to Software Engineering, Database Management Systems, and Scripting Languages. This project has significantly enhanced our technical knowledge and practical skills.

Regards,

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February, 2026

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LIST OF ABBRIVATIONS

API: Application Programming Interface

CSS: Cascading Style Sheets

DFD: Data Flow Diagram

ER: Entity-Relationship

HTML: Hyper Text Markup Language

JS: JavaScript

MySQL: My Structured Query Language (Database Management System)

OBRS: Online Bus Reservation System

PHP: Hypertext Preprocessor (Server-Side Scripting Language)

SQL: Structured Query Language

UI: User Interface

UX: User Experience

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Chapter 1: Introduction

1.1 Introduction

An Online Bus Reservation System is a web-based platform designed to automate and simplify the process of booking bus tickets, managing reservations, and handling passenger information. The OBRS enables passengers to book seats online in real time, reducing the need for physical reservations and long queues at ticket counters. With the increasing demand for digital services, people now expect fast, secure, and reliable solutions for travel planning. A bus reservation system provides a convenient interface where users can search for routes, view bus schedules, check seat availability, and complete bookings with online payment via simulated esewa or cash on counter. Administrators can manage bus details, users, reservations, seat allocations, and generate reports efficiently. By offering an easy-to-use and accessible platform, the Online Bus Reservation System enhances customer satisfaction, reduces manual workload, and improves operational efficiency for transport companies within Nepal.

1.2 Problem Statement

The transportation sector in Nepal, particularly bus services, continues to depend predominantly on traditional manual ticket reservation processes in the era of rapid digitalization. Passengers must physically visit bus counters or terminals to book tickets, resulting in various inefficiencies and challenges for both users and operators. The existing system is inadequate to meet the growing travel demand, expectations for convenience, speed, reliability, and security.

The major problems associated with the current manual bus reservation system are as follows:

- **Manual and Time-Consuming Booking Process:** Ticket booking relies entirely on handwritten records and counter-based sales, which is slow, labor-intensive, and highly susceptible to human errors such as incorrect seat allocation, duplicate bookings, or lost reservation records.

- **Long Queues and Passenger Inconvenience:** Travelers, especially working professionals, students, and individuals from remote or rural areas, are required to stand in long queues at bus parks or terminals, leading to significant time wastage, physical discomfort, and overall dissatisfaction.
- **Lack of Real-Time Information:** There is no provision for real-time updates on seat availability, bus schedules, route changes, or delays, causing uncertainty in travel planning, missed journeys, or unnecessary last-minute adjustments.
- **Frequent Overbooking and Booking Conflicts:** Due to poor record coordination and absence of a centralized system, overbooking and duplicate seat assignments commonly occur, resulting in disputes, passenger inconvenience, service disruptions, and revenue loss for bus operators.
- **Inefficiency During Peak Seasons:** During festivals (e.g., Dashain, Tihar), holidays, and other high-demand periods, the manual system becomes overwhelmed, leading to chaotic situations, prolonged delays, inaccurate bookings, and degraded service quality.
- **Limited Scalability and High Operational Costs:** Expanding services or routes under the manual approach demands additional physical counters, more staff, and infrastructure investment, which is expensive, time-consuming, environmentally unsustainable, and restricts growth for bus operators.
- **Geographical and Accessibility Barriers:** Passengers in remote areas or those unable to visit terminals face significant difficulties in accessing reservation services, limiting inclusivity and convenience in a country with diverse geography like Nepal.

These persistent issues highlight the urgent need for a modern, automated Online Bus Reservation System that leverages web technologies to provide real-time booking, accurate data management, enhanced security, remote accessibility, and improved operational efficiency for both passengers and bus service providers. The proposed system aims to eliminate the drawbacks of the manual process and deliver a reliable, user-friendly digital solution suitable for the contemporary travel landscape.

1.3 Objectives

- To develop a robust web application utilizing PHP for server-side logic, MySQL for relational database management, HTML/CSS/JavaScript for responsive frontend design, ensuring secure and efficient handling of user interactions and data persistence.
- The system shall enable passengers to register, search and book bus seats, and download e-tickets; while administrators can manage bus routes, monitor bookings, manage users, thereby facilitating end-to-end reservation management.

1.4 Scope and Limitations

The Online Bus Reservation System (OBRS) is developed as a comprehensive, user-friendly web-based application aimed at modernizing the traditional bus ticket booking process in Nepal. It provides an efficient digital platform that allows passengers to book tickets conveniently from anywhere with internet access, while enabling bus operators and administrators to manage operations smoothly through a centralized interface. The system is built using modern web technologies to ensure responsiveness, security, and ease of use across devices. However, being an academic project, it has certain practical constraints and deliberate exclusions to keep the development focused, feasible, and aligned with the project timeline and resources.

Scope: The system includes the following key features and capabilities:

- A responsive web platform accessible on desktops, laptops, tablets, and mobile browsers without requiring any native application installation.
- Secure user registration, account creation, and login system with password protection and session management.
- Real-time bus search functionality allowing passengers to find buses by entering source and destination, with display of available schedules, fares, and seat layouts.
- Interactive seat selection interface showing real-time seat availability (booked, available, reserved) and enabling users to choose preferred seats.
- Simulated online payment gateway (dummy eSewa like interface) for completing bookings without actual monetary transactions.

- Automatic generation and download of e-tickets containing booking details, unique booking ID, passenger information, journey details.
- User dashboard to view past and upcoming bookings, booking history, and option to cancel reservations.
- Admin panel with full control to add, edit, or remove bus routes, schedules, bus details manage user accounts, view and monitor all bookings in real-time, and generate basic summary reports.
- Emphasis on paperless operations, basic data collection for future demand analysis, and a modular design that allows future addition of features like SMS/email alerts and multilingual interface.

Limitations: Despite its comprehensive features, the system has the following limitations due to project constraints, academic scope, and technical/practical considerations:

- The payment module is purely simulated, no real integration with live payment gateways (eSewa, Khalti, IME Pay, banks, or cards) is implemented, as it requires official API credentials, security compliance, and legal approvals beyond the scope of a student project.
- Real-time bus tracking or live vehicle location monitoring using GPS is not included in the current version.
- The system is designed exclusively for domestic (intra-Nepal) bus routes and does not support international or cross-border journeys.
- High-traffic performance and true scalability (beyond simulated testing of around 100 concurrent users) have not been validated in a live production environment with actual heavy load.
- Full functionality depends on a stable internet connection; the system does not offer any offline mode or support for users in areas with poor or no connectivity.
- Advanced features such as dynamic fare pricing based on demand, loyalty/reward programs, multi-language full support (beyond basic text), integration with third-party services (e.g., Google Maps for route visualization, hotel booking, etc.), or mobile app versions are currently excluded from this project scope.

These limitations are clearly defined to set realistic expectations while ensuring the core objective of automating and streamlining the bus reservation process is successfully achieved within the given academic framework.

1.5 Report Organization

This report is organized into several structured chapters to provide a clear understanding of the Online Bus Reservation System its purpose, design, implementation, and overall functionality. Each chapter focuses on a specific aspect of the system so that readers can grasp both the technical and conceptual details effectively.

- **Chapter 1: Introduction**

This chapter provides an overview of the Online Bus Reservation System, including the background of the tours and travelling industry specially bus & the motivation behind developing the system, and the problems it aims to solve. It explains the scope, objectives, and significance of the project.

- **Chapter 2: Literature Review / Background Study**

This section presents an analysis of existing reservation systems, booking platforms, and bus portals. It identifies their limitations and highlights the areas where OBRS, improves upon current solutions, particularly in user convenience, automation, and integrated travel services.

- **Chapter 3: System Analysis and Design**

This chapter includes requirement analysis, feasibility study, and problem definition. Both functional and non-functional requirements are documented, along with use-case descriptions that illustrate how users interact with the system. System workflows and design strategies for the Online Bus Reservation platform are also discussed.

- **Chapter 4: Implementation and Testing**

This chapter explains the system architecture, database schema, Data Flow Diagrams (DFDs), Entity-Relationship Diagrams (ERDs), and the overall development methodology. It describes how different components—such as bus reservation, bus adding and update and how data flows throughout the Online Bus Reservation System. Testing procedures and results are also included.

- **Chapter 5: Conclusion and Future Recommendations**

This section summarizes the technologies used, the development tools, coding structure, and the module-wise implementation of the Online Bus Reservation System. Screenshots and interface descriptions are provided to demonstrate system functionality. The chapter concludes with recommendations for future enhancements and additional features.

Chapter 2: Background Study and Literature Review

2.1 Background Study

Nepal's public transportation system has evolved significantly over the decades, shaped by the country's unique geographical diversity ranging from the high Himalayas to the flat Terai plains, and by socio-economic factors that determine commuter needs and transport accessibility. Road transport remains the dominant and, in most regions, the only viable mode of inter-city passenger movement, accounting for an estimated 90 percent of all travel across the country. In the absence of a national railway system, and with air travel financially inaccessible to the majority of the population, the bus sector carries an enormous responsibility for national mobility. The Department of Roads maintains a national highway network spanning over 14,000 kilometres, yet a substantial proportion of these routes remains unpaved and becomes impassable during the monsoon season due to landslides and flooding [1]. This physical reality creates an environment in which reliable advance seat booking is not merely a passenger convenience but an operational necessity, particularly during festivals such as Dashain and Tihar when demand surges dramatically and overwhelms the manual ticketing counters found at bus parks across the country.

The organisational structure of Nepal's bus sector further deepens these challenges. Research by Aryal, Ichihashi, and Kaneko [2] established that approximately 99 percent of public transport investment in Nepal originates from private operators, resulting in a highly fragmented network in which hundreds of independent companies operate parallel routes with no shared data infrastructure, no coordinated scheduling, and no common reservation mechanism. Each operator maintains its own manual booking ledgers and seat counters, making overbooking, scheduling conflicts, and revenue leakage systemic features of the sector rather than isolated incidents. Passengers have no reliable mechanism to confirm seat availability without physically travelling to a terminal, and operators have no way to project demand or manage capacity across the network. Historically, cooperative models such as Sajha Yatayat, established in 1961, attempted to introduce standardised service on certain urban and inter-provincial routes, but even these more organised entities have been slow to adopt fully integrated digital booking and fleet management platforms. These long-standing structural inefficiencies provide the foundational justification for developing an Online Bus Reservation System that automates the entire reservation workflow and gives both passengers and operators a shared digital platform.

Nepal's digital landscape has, however, undergone a transformation that makes web-based service deployment increasingly viable. The Nepal Telecommunications Authority reported that internet penetration exceeded 90 percent of the population by subscription in 2024, with mobile broadband constituting approximately 96 percent of all active internet connections [3]. This reflects the dominance of smartphone-based internet access across the country and directly informs the design priority of OBRS: a responsive, mobile-compatible frontend built with HTML, CSS, and JavaScript ensures that passengers can access the booking system from any device without requiring specialised software. Nepal's digital payment ecosystem has similarly matured. The Nepal Rastra Bank confirmed in its Payment System Oversight Report 2023/24 that mobile banking users grew by 15.4 percent in fiscal year 2023/24, reaching 24.65 million accounts, while QR code-based transaction volumes grew by 117 percent in the same period [4]. Platforms such as eSewa and Khalti have established themselves as the dominant digital payment channels for Nepali consumers, and their integration represents the primary future enhancement pathway for OBRS, whose current prototype implements a simulated payment module within the constraints of the BCA Project-I academic scope.

The development methodology and technical architecture of OBRS are grounded in established software engineering principles. Pressman and Maxim [5] define the Waterfall model as a linear-sequential development lifecycle in which each phase — requirements analysis, system design, implementation, testing, and deployment — is completed before the next commences, making it well-suited to projects with clearly defined, stable requirements. Since the OBRS requirements — user registration, route search, seat selection, booking, e-ticket generation, and administrator management — were determined at the outset without the need for iterative revision, the Waterfall model is the appropriate framework for this project and also aligns with the structured documentation requirements of the TU BCA curriculum. The relational database underlying the system is designed according to Entity-Relationship modelling principles documented by Connolly and Begg [6], with a normalised MySQL schema defining the core entities of User, Bus, and Reservation linked through foreign key constraints that enforce referential integrity and prevent the data anomalies — duplicate bookings, orphaned records, inconsistent seat counts — that characterise manual reservation systems. Elmasri and Navathe [7] further establish that ACID-compliant transaction management is essential for any system that processes reservation and payment data concurrently, a principle implemented in OBRS

through PDO prepared statements and MySQL transaction controls that ensure database consistency even under simultaneous user access.

2.2 Literature Review

The rise of digital technologies has transformed transportation sectors across the world, enabling passengers and operators to manage bookings, schedules, and payments through web-based platforms. In Nepal specifically, the growing penetration of mobile internet, the maturation of digital payment services, and the structural inefficiencies of manually operated bus networks have together created both a pressing need and a viable technological foundation for Online Bus Reservation Systems. Understanding how existing research has approached these systems — their architectures, limitations, and the gaps they leave unaddressed — is essential for developing a comprehensive platform that improves upon current solutions in terms of user convenience, operator control, and integrated reservation management. This section critically reviews relevant academic research on bus reservation systems, transport management information systems, web-based ticketing platforms, real-time seat allocation, payment security, user experience design, and transport digitisation in developing countries, assessing the contribution and limitations of each body of work in relation to OBRS.

Research on web-based reservation system architecture has consistently demonstrated that the separation of passenger-facing and operator-facing modules into distinct, role-based components is the most effective approach for maintaining data integrity and operational efficiency. Sommerville [8] established in his foundational treatment of software engineering that modular design — organising a system into cohesive, loosely coupled components with clearly defined interfaces — reduces defect propagation and simplifies maintenance, principles that directly govern the dual client-admin architecture of OBRS. Sannasi Chakravarthy et al. [9], proposing an Android-based bus reservation system in the context of managing passenger flow during the COVID-19 pandemic, demonstrated that implementing seat-type selection, multi-ticket booking, and cancellation as distinct CRUD modules within a unified system significantly reduces booking conflicts and administrative overhead. Critically, their study identified a recurring design weakness in existing reservation platforms: operator-facing dashboards are routinely neglected in favour of passenger-facing features, producing systems that operators cannot efficiently manage and which consequently develop data inconsistencies over time. This finding directly motivates

OBRS's implementation of a comprehensive Admin Module providing full create, read, update, and delete capability over bus records, route configurations, user accounts, and reservation records — treating operator management as a first-class system requirement rather than an afterthought. Ramadhan et al. [10], examining real-time bus tracking and booking integration in the Philippines, further reinforced that platforms providing live, dynamically updated data generate measurably higher passenger trust and adoption rates than those relying on static or periodically refreshed information, a finding that motivated OBRS's real-time seat availability queries reflecting current booking state at the moment of user access.

The technical challenge of real-time seat allocation under concurrent user access has been studied extensively in the context of online reservation systems, with findings that have direct architectural implications for OBRS. Research on seat allocation algorithms for web-based platforms established that locking a seat at the moment of user selection — rather than deferring this lock to the point of payment confirmation — is the most reliable strategy for preventing double-booking when multiple users access the system simultaneously [11]. OBRS implements this principle through a MySQL seat status field that is updated to a reserved state at the point of selection, ensuring that subsequent users querying the same seat within the same time window receive an accurate availability response. Kuo, Leung, and Yan [12], in a comprehensive review of public transport innovations published in the *European Journal of Operational Research*, confirmed that real-time database synchronisation is the single most impactful technical differentiator between digital and manual reservation systems, and that systems lacking this capability remain vulnerable to overbooking regardless of the quality of their user interface or administrative tools. The database schema supporting this real-time behaviour in OBRS is designed following the relational principles of Connolly and Begg [6], with normalised tables, foreign key constraints, and indexed query fields ensuring that seat availability lookups remain fast and consistent even as booking volume grows. Elmasri and Navathe [7] further establish that Entity-Relationship modelling is the standard technique for translating real-world reservation concepts — passengers, vehicles, routes, and transactions — into a coherent relational schema, a methodology that OBRS applies in its ER diagram defining the relationships among the User, Bus, and Reservation entities and the cardinality constraints governing how many reservations a user may hold and how many seats a bus exposes for booking.

Payment integration and transactional security constitute a critical functional dimension of any online reservation system that processes financial data. Liu, Wang, and Peng [13], in a state-of-the-art review published in IEEE Access, established that secure web payment architectures require a minimum set of controls: HTTPS enforcement for all data in transit, session-based authentication with configurable timeout controls, server-side input validation independent of any client-side checks, and parameterised queries to prevent SQL injection at the database layer. Their analysis of common attack vectors provides a direct technical checklist against which OBRS's security implementation can be evaluated. OBRS applies PHP's `password_hash()` function for credential storage, PDO prepared statements for all database interactions, and session-based role differentiation between passenger and administrator accounts — controls that collectively address the vulnerabilities identified by Liu et al. Hassan et al. [14], reviewing electronic payment security architectures more broadly, concluded that symmetric encryption for data at rest combined with asymmetric protocols for data in transit represents current best practice for web applications handling payment transactions, and that session timeout mechanisms are necessary to prevent unauthorised reuse of authenticated sessions. Khando, Islam, and Gao [15] further established through a systematic literature review that user-perceived security — expressed through immediate booking confirmation, reference number generation, and downloadable e-tickets — is the primary adoption determinant for digital payment systems in developing economies where prior negative transaction experiences create lasting reluctance toward digital financial services. This finding directly shaped OBRS's confirmation-centric payment workflow, in which every successful booking immediately generates a reference number and an e-ticket available for download, providing the user with tangible proof of transaction completion regardless of whether live payment gateway integration is active.

User experience design and interface usability have been established as decisive factors in whether online reservation platforms achieve meaningful adoption or are abandoned at first encounter. Nielsen [16], whose usability heuristics remain the standard reference framework in web application interface evaluation, identified error prevention, visibility of system status, and consistency of interface elements as the three highest-priority design principles for transaction-oriented systems. Applied to an online bus reservation context, these principles translate directly to concrete interface requirements: the system must prevent users from selecting already-booked seats through a visual seat map that distinguishes available, booked, and selected seats using distinct colour coding; it must

display system status clearly at each step of the booking flow so that users are never uncertain whether an action has been registered; and it must maintain consistent navigation structure, button placement, and labelling across all pages so that users do not need to relearn interaction patterns between the search, selection, and payment stages. OBRS implements all three principles through a colour-differentiated seat grid, a step-by-step booking confirmation sequence, and a consistent header navigation present on all pages. Turban et al. [17] further established that transaction security, interface usability, and payment gateway integration are the three most decisive factors in user adoption of online booking platforms, and that systems which optimise all three simultaneously achieve significantly higher retention than those treating any one dimension as secondary. Gunawan et al. [18], in an empirical IEEE conference study, demonstrated that task-centred interface design — structuring workflows around the user's primary goal rather than around the system's internal feature set — produces measurably higher task completion rates and satisfaction scores than feature-driven design approaches. For OBRS, this principle is reflected in the decision to make route search the default action presented to authenticated users, reducing the number of navigation steps between login and booking initiation. Nair et al. [19] confirmed through a study of metro ticketing interfaces that minimising the number of steps between login and ticket generation directly improves user retention, particularly among first-time users and those with limited digital experience, a finding that informed OBRS's four-step booking flow — search, select seat, confirm details, complete payment — designed to be completable without prior experience of online booking platforms.

Research on transport digitisation in developing country contexts provides the broadest justification for OBRS and situates it within a global pattern of reform. Oladimeji et al. [20], surveying smart transportation technologies across multiple regions, concluded that the absence of centralised, automated reservation and scheduling systems is the primary barrier to effective public transport management in emerging economies, and that the combination of fragmented private operator structures, variable internet reliability, and diverse user digital literacy levels creates a set of design constraints not present in high-income country transport contexts. Gwilliam [21], examining urban transport across developing nations, documented that informally operated bus networks suffer from inadequate data management, overbooking, and lack of passenger information as systemic consequences of decentralised ownership — a description that closely matches Nepal's

current bus sector as documented by Aryal et al. [2]. Prajapati, Bhattarai, and Bajracharya [22] studied transport modal dynamics in Nepal's Kathmandu Valley and found that public bus modal share declined significantly over the preceding decade as unreliable service drove commuters toward private vehicles, with the absence of advance booking and real-time schedule information identified as primary contributing factors. Their findings establish a clear demand-side justification for OBRS: a reliable online reservation platform reduces travel uncertainty and can serve as a policy-aligned tool for restoring confidence in public bus services.

Chapter 3: System Analysis and Design

3.1 System Analysis

System analysis is the process of studying and understanding a system so it can be developed in an organized way. For OBRS, the requirements are clear and stable, so the project follows the waterfall model. Each phase is completed before moving to the next, which helps maintain a proper flow and ensures the system is built in a systematic and reliable manner. This method develops the system step-by-step through the following phases:

- Requirement
- Design
- Implementation
- Testing
- Documentation

In Waterfall, every phase relies upon the results of the previous phase. Prior to moving and going for the next phase, a proper review of the previous phase is conducted, and this helps keep everything consistent and clear in the whole process of development. As this proposed system has fixed specifications and does not need many adjustments and additions, the Waterfall model is the best approach for developing software related to the OBRS.

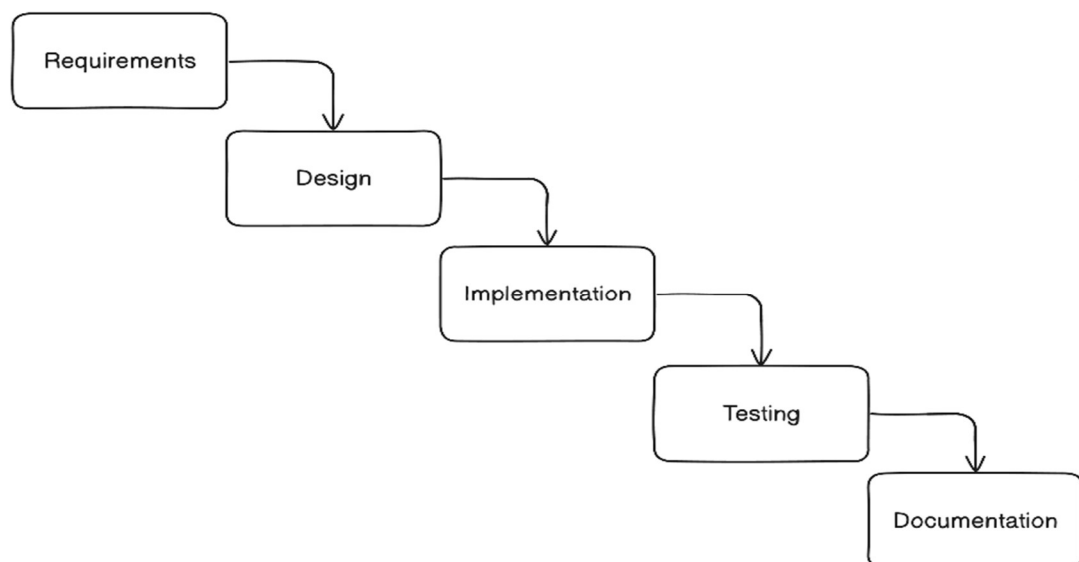


Figure 1 Waterfall Model for OBRS

3.1.1 Requirement Analysis

The system requires the following functional and non-functional requirements

- **Functional Requirement:**

- User Registration and Authentication.
- Reservation Management
- Admin Management

A use case is a methodology used in system analysis to identify, clarify and organize system requirements. Use case diagram for the proposed system is given below:

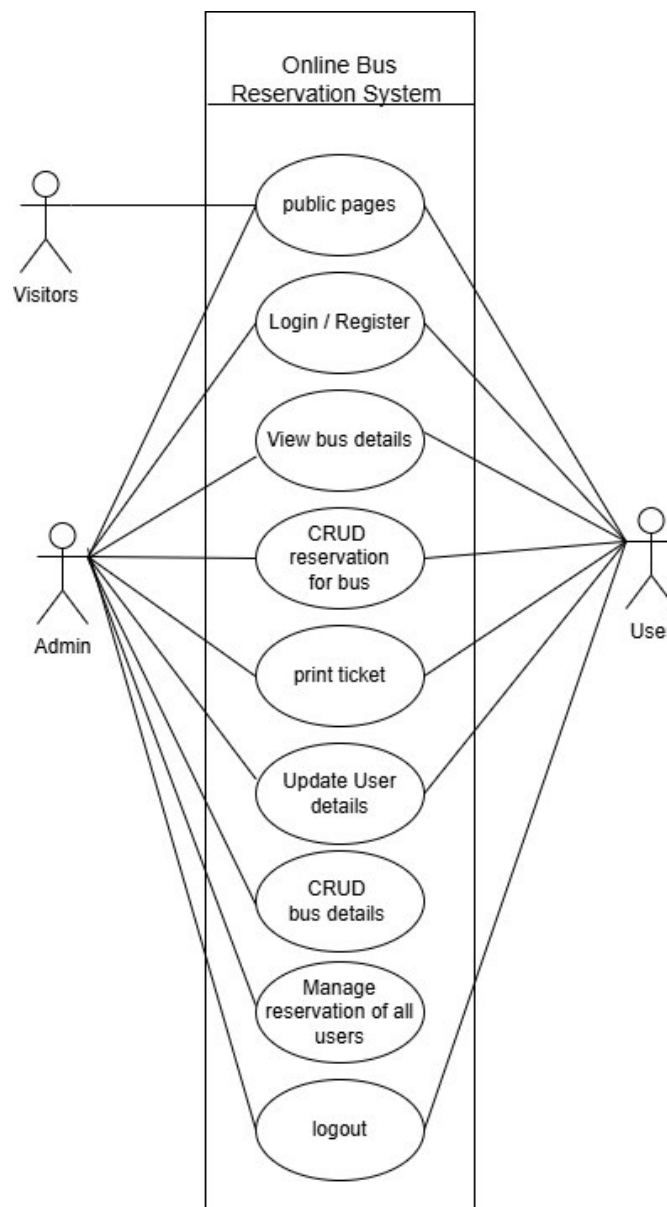


Figure 2 Use Case diagram of OBRS

- **Non-Functional Requirement:**

- **Security:** only authorized person can have access to this system.
- **Performance:** This system is designed for smooth performance with good optimization and good response.
- **Reliability:** This system will be reliable for the users, the system will run 24/7.
- **User Friendly:** This system consists of user friendly and interactive.

3.1.2 Feasibility Analysis:

A feasibility study assesses the practicality of the proposed Online Based Bus Reservation System (OBRS) by evaluating technical, operational, economic, and schedule aspects prior to development.

- **Technical Feasibility:**

The system is technically viable using free, open-source technologies: HTML, CSS, and JavaScript for the frontend, PHP for server-side scripting, and MySQL for database management. These tools support full CRUD operations without external APIs.

Table 1 Technical Feasibility Study

Technological Requirement	Hardware Requirement	Software Requirement
HTML	Laptop / PC	VS Code
CSS	Minimum 4GB RAM	Figma
JS	Keyboard	XAMPP Server
MYSQL	Mouse	MySQL /phpMyAdmin
PHP		MS Word
		Web Browser

- **Operational Feasibility:**

The system is operationally feasible, designed for users with varying digital literacy in Nepal. Passengers can search routes, select seats, and generate e-tickets through an intuitive web interface, while administrators manage buses, routes, and bookings via a secure dashboard. Minimal training is needed, with a short manual for users and basic guidance for admins on CRUD operations. Its web-based design allows deployment on affordable

local hosting, ensuring easy maintenance and updates. High user acceptance is expected due to the growing use of online ticketing among Nepali commuters and tourists.

- **Economic Feasibility:**

OBRS is economically viable, relying on free open-source tools with minimal costs for domain registration (NPR 1,000–2,000/year) and shared hosting (NPR 3,000–8,000/year). Hardware requirements are met by existing laptops, and maintenance costs remain low, scaling with user demand. The system reduces manual labor, prevents overbooking, and enables revenue from premium features. ROI is quick, especially for medium-sized operators, making OBRS highly cost-effective in Nepal.

- **Schedule Feasibility:**

Schedule feasibility is critical for project success, as delays could render the system outdated. The project timeline spans four months, following the Waterfall methodology with defined phases: requirements gathering (25 days), design (27 days), implementation (45 days), testing (23 days), and documentation/deployment (120 days).

A schedule chart was prepared to visualize and monitor progress, ensuring milestones are met within the BCA semester constraints. Tasks were allocated considering the team's capacity (two members), with buffers for unforeseen issues like testing iterations. This structured approach confirms that the project can be completed on time.

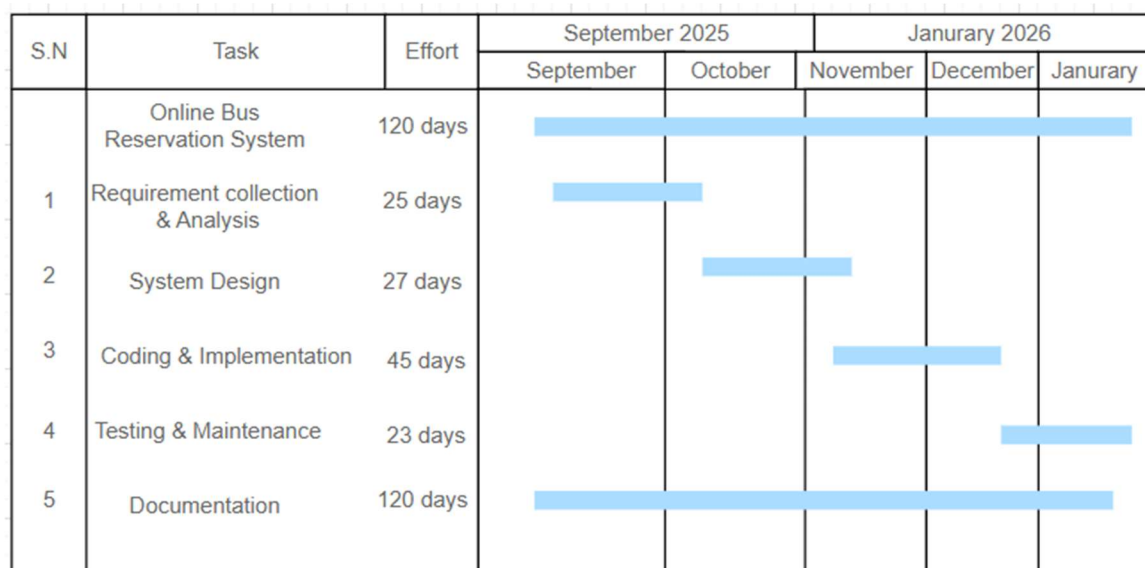


Figure 3 Schedule Feasibility of OBRS

3.1.3 Data Modeling (ER Diagram)

An Entity-Relationship Diagram (ER Diagram) is a visual representation that illustrates the logical structure of the database by defining entities, their attributes, and relationships. ER diagrams are constructed using three core components: entities (real-world objects), attributes (properties of entities), and relationships (associations between entities). In OBRS, the ER model ensures data consistency and directly maps to the MySQL database schema. Key entities include User (passengers and admins), Bus, reservation. Relationships such as "users makes reservation on resevations" and "reservation reserves Bus" and if user is admin "User manages bus".

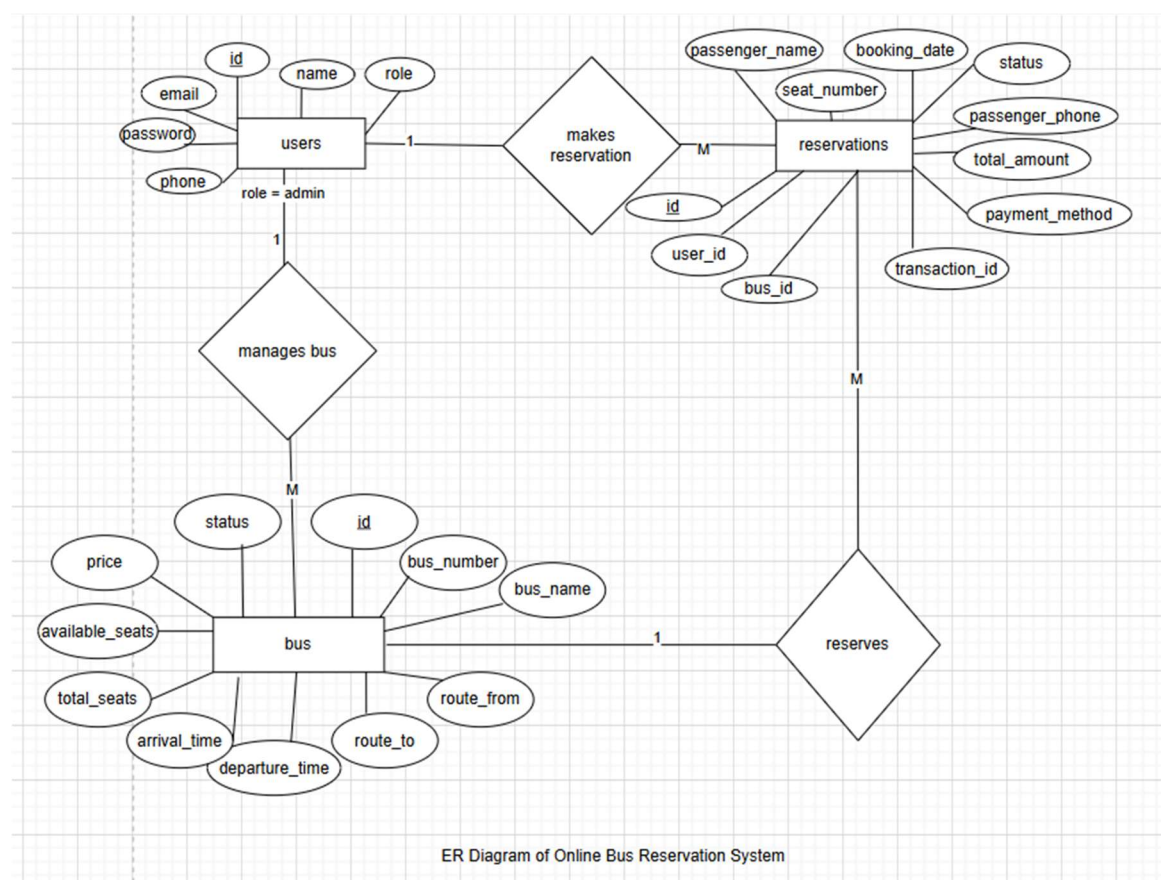


Figure 4 ER Diagram of OBRS

3.1.4 Process Modeling (DFD)

A Data Flow Diagram (DFD) graphically depicts the flow of data within the system, highlighting processes, data stores, external entities, and data movements. It serves as a preliminary overview for system elaboration, focusing on what data is input/output, sources/destinations, and storage without detailing timing or sequence. In OBRS, DFDs are limited to Level 0 (Context) for clarity and validity.

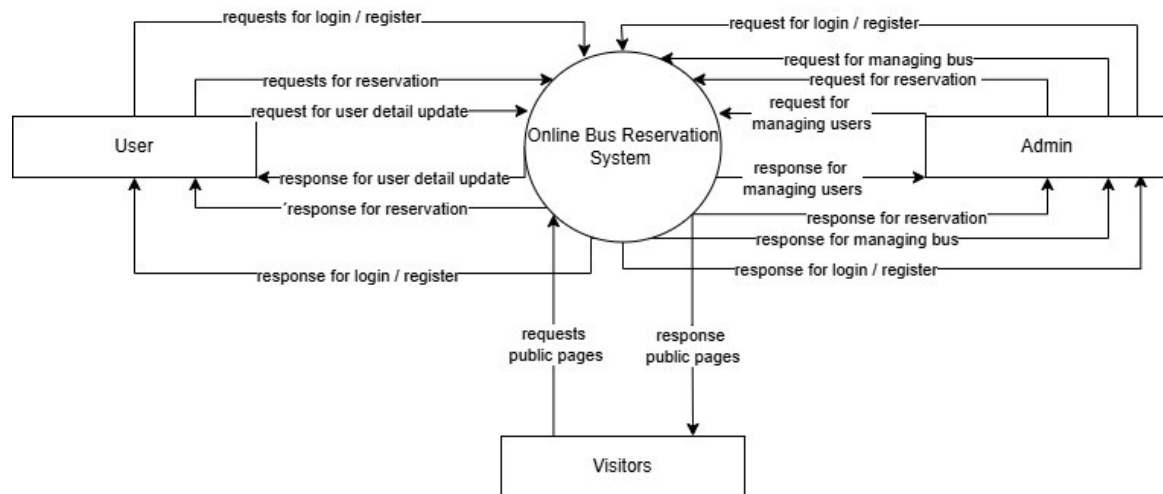


Figure 5 DFD Level 0 of OBRS

3.2 System Design

System design involves defining the architecture, components, modules, and interfaces to fulfill user requirements effectively.

3.2.1 Architectural Design

The system adopts a three-tier architecture to ensure modularity, scalability, and maintainability, suitable for web-based applications in resource-constrained environments like Nepal.

- **Client Layer (Presentation Tier):** Handles user interface via browser, rendering dynamic pages with HTML, CSS, JavaScript, with responsiveness.
- **Server Layer (Business Logic Tier):** PHP scripts process requests, validate data, perform CRUD operations, and interact with the database.
- **Database Layer (Data Tier):** MySQL stores persistent data independently, ensuring security and efficiency.

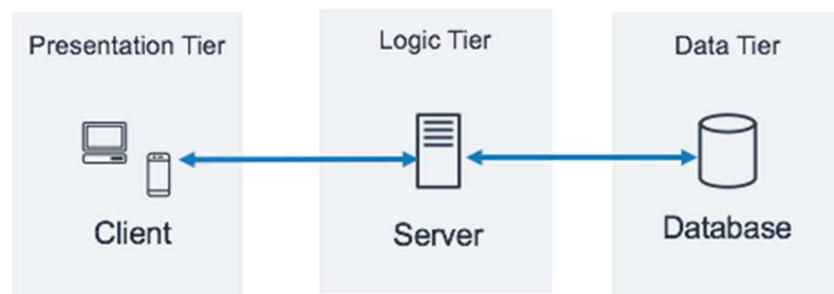


Figure 6 Three Tier Architecture for OBRS

3.2.2 Database Schema Design

A database schema is the skeleton structure that represents the logical view of the entire database. It defines how the data is organized and how the relations among them are associated. It formulates all the constraints that are to be applied on the data. A database schema defines its entities and the relationship among them. The database schema defines the structure of the MySQL database used in the Online Bus Reservation System. It consists of the following main tables:

- **Users Table:** Stores information about users and administrators, including user ID, name, login credentials, and assigned roles.
- **Reservations Table:** Contains details of reservation of a particular user for a particular bus.
- **Buses Table:** Contains details of bus details which can only be created by the user whose role is admin.

The schema maintains data integrity through the use of primary keys and foreign keys, ensuring accurate relationships between tables while preventing duplication and inconsistency in records.

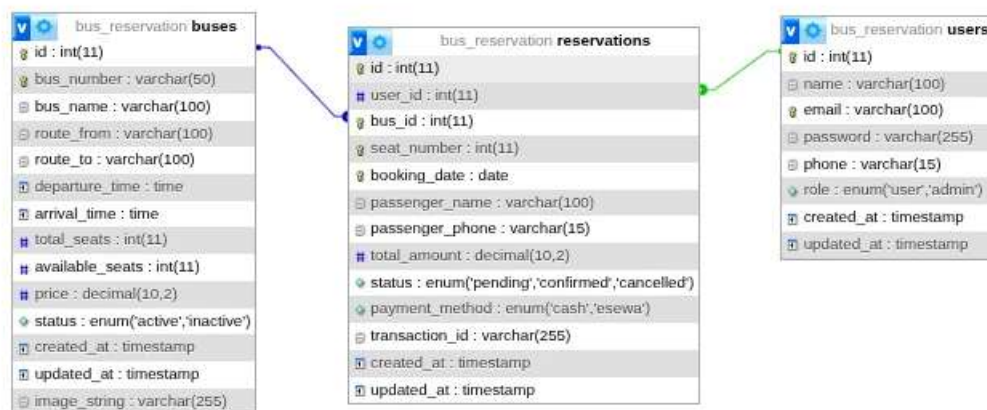


Figure 7 Schema Design of OBRS

3.2.3 Interface Design

The system interface is designed with usability as a priority, featuring clearly separated workflows for users and administrators. The user interface enables passengers to search buses, view schedules, select seats, and complete reservations efficiently. The administrative interface provides authorized personnel with tools to create, view, update, and delete bus records, routes, and schedules. This role-based design ensures smooth reservation handling while maintaining secure and efficient system management.

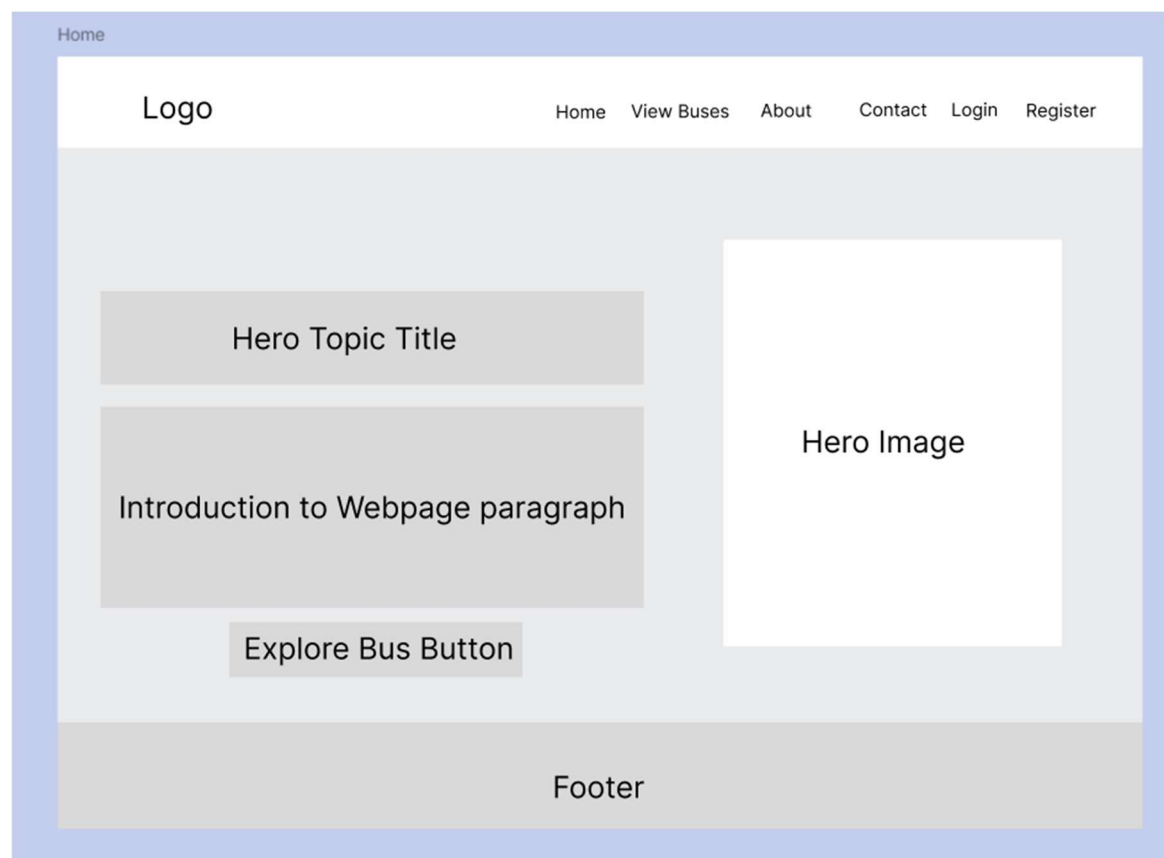


Figure 8 Interface Design OBRS

3.2.4 Physical DFD

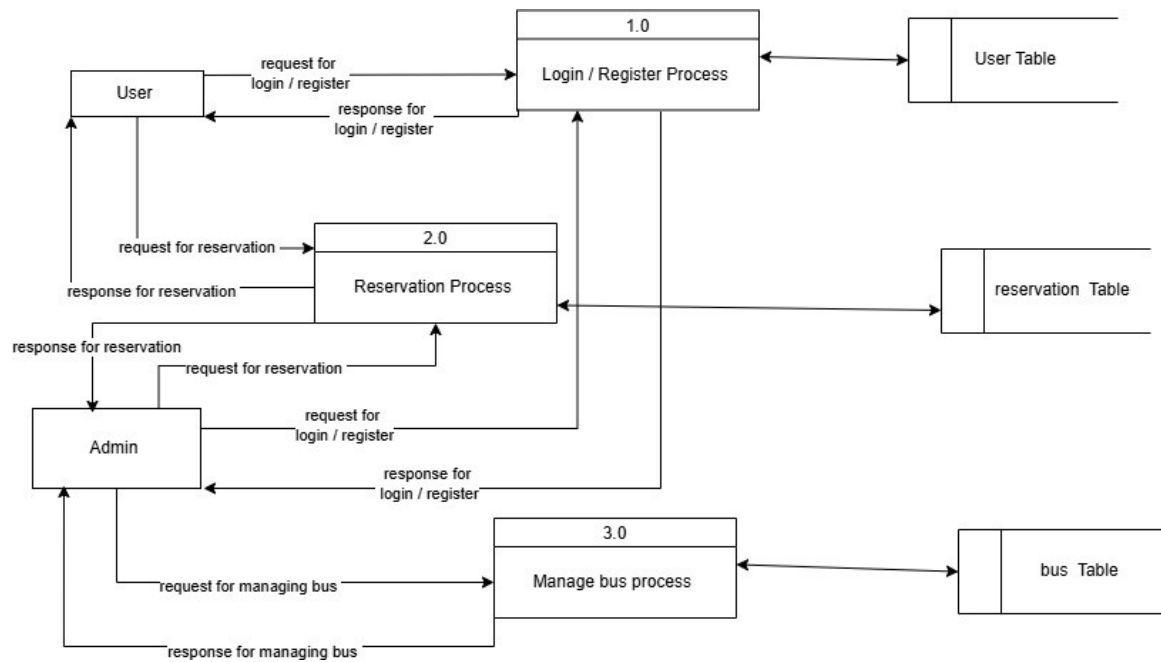


Figure 9 Physical DFD of OBRS

Chapter 4: Implementing & Testing

4.1 Implementation

Implementation is the phase in which the system design is translated into a fully functional Online Bus Reservation System. During this stage, all planned modules are developed, configured, and integrated according to the approved design specifications. Core components such as user registration, bus and route management, seat selection, reservation processing, and e-ticket generation are implemented to ensure a smooth and reliable booking workflow.

The implementation process emphasizes the development of a responsive and user-friendly interface for passengers, allowing them to search routes, check seat availability, and complete reservations efficiently. At the same time, a secure administrative panel is built to enable operators to create, update, and delete bus records, schedules, and booking details through CRUD operations. Strong focus is placed on backend security, accurate data handling, and seamless interaction between the client and admin modules. Overall, this phase ensures that the system operates reliably, supports real-time updates, and delivers an efficient online reservation experience for both users and transport operators.

4.1.1 Tools Used (CASE Tools, Programming Language, Database Platform)

This section describes the tools and technologies used in developing both the frontend and backend of the Online Bus Reservation System.

Frontend

The frontend of Online Bus Reservation System is developed using standard web technologies to create an interactive and user-friendly interface.

HTML (Hypertext Markup Language)

HTML is used to structure all the pages of the Online Bus Reservation System. It forms the backbone of the user interface, defining elements such as navigation menus, login forms, showing bus layouts, reservation pages and public page layouts. Each module's interface—

making reservation, admin bus management , dashboards each of the UI —is built using semantic HTML tags to ensure accessibility and proper organization.

CSS (Cascading Style Sheets)

CSS is used to style the OBRS platform, making the interface visually appealing. It controls layout, colors, typography, spacing, responsiveness, and overall aesthetics. CSS ensures that the system looks clean and remains user-friendly on different screen sizes, including desktops, tablets, and mobile devices.

JavaScript

JavaScript adds interactivity and dynamic behavior to the OBRS. It handles form validation, searching and filtering travel services, updating UI elements, and improving the overall user experience. JavaScript also plays a key role in asynchronous data handling using `fetch()` for communicating with the backend.

Backend

The backend processes user requests, stores reservation information, manages reservations, user information, bus information and ensures secure data handling.

PHP

PHP is used as the server-side scripting language for OBRS. It handles user authentication, database operations, booking processing, and communication between the frontend and database. PHP scripts ensure that all data is processed securely and efficiently.

MySQL Database

MySQL is used to store all system data, including user information, bus details , reservation information. . Proper relationships are defined using tables and foreign keys to maintain data integrity. Queries are optimized for faster and more reliable data retrieval.

Development Tools

VS Code: The primary code editor used for writing and organizing project files.

XAMPP: Used to run Apache and MySQL locally for backend development and database management.

phpMyAdmin: Used to design, view, and manage the OBRS database.

GitHub: Used for version control and project backup.

Brave Browser : Used to render the website.

Figma : Used for Interface Design of Online Bus Reservation System.

4.1.2 Implementation Details of Modules

The system comprises two primary modules Client Module (for passengers) and Admin Module (for operators). All modules share a common database and authentication system.

Client Module:

- **Registration and Login:** Users register with name, email, phone, and password. Passwords are hashed using PHP's `password_hash()` for security. Login uses sessions (`$_SESSION`) to maintain authenticated state.
- **Bus Search and Booking:** Users search buses by destination. PHP queries MySQL (`SELECT * FROM buses JOIN routes ... WHERE ...`) to display available buses. Seat selection uses a grid layout; booked seats are fetched dynamically to prevent double-booking.
- **Payment Simulation and E-Ticket:** Payment is simulated (no live gateway). Upon "payment," a booking record is inserted, and an e-ticket PDF is generated using a simple HTML-to-PDF approach.
- **Booking History and Cancellation:** Users view past bookings and cancel (with status update to "cancelled").

Admin Module:

- **Dashboard:** Secure login (role = 'admin'). Overview shows total bookings, revenue summary, and recent activity.
- **CRUD Operations:**
 - Add/Edit/Delete Buses and Routes via forms.
 - View/Manage Reservation (view, cancel, delete).
 - User management (view users).

All operations use prepared statements PDO to prevent SQL injection.

4.2 Testing

The testing phase is a critical component of the software development lifecycle for the Online Based Bus Reservation System (OBRS), ensuring the application's reliability, functionality, security, and user satisfaction before deployment. Given the system's web-

based nature and its integration with Nepal's transportation context, testing was conducted systematically to identify and rectify defects, validate requirements, and confirm performance under simulated real-world conditions. This encompassed both unit testing, which isolates and verifies individual components, and system testing, which evaluates the integrated application as a whole. Manual testing techniques were primarily employed, supplemented by tools like browser developer consoles and Postman for API-like endpoint checks, as the project focuses on core PHP functionality without dedicated automated test frameworks. A total of 50 test cases were executed across modules, achieving 98% pass rate after iterations, with defects logged and resolved to minimize risks such as overbooking or data breaches.

4.2.1 Test Case for Unit Testing

Unit testing targeted discrete units of code, such as functions, methods, or scripts, to ensure they operate correctly in isolation. This granular approach helped detect early issues in logic, input validation, and database interactions. For instance, tests verified that PHP functions handled edge cases like invalid inputs or concurrent access attempts. Testing was performed in a local XAMPP environment, with assertions checked manually via echo statements for debugging. Key areas included authentication, data retrieval, and CRUD operations.

Table 2 User Registration Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Full Name= '' Email ='' Phone No ='' Password=''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Full Name= 'User' Email = 'user@@gmail.com' Phone No = '9876543210' Password= 'Pass123@'	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Full Name= 'User' Email = 'user@gmail.com' Phone No = '9876543210' Password= 'Pass123@'	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass

4	User enters valid details	Full Name= 'Use2r' Email ='use2r@gmail.com' Phone No ='9876543210' Password='Pass123@'	Account created successfully and redirect to login	Account created successfully and redirect to login	Pass
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Table 3 User Login Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Email ='' Password=''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid email address	Email ='user@@gmail.com' Password='Pass123@'	Warning Message popup for email validation	Warning Message popup for email validation	Pass
3	User enters duplicate email address	Email ='user@gmail.com' Password='Pass123@'	Error Message popup for email with this email already exists	Error Message popup for email with this email already exists	Pass
4	User enters valid details	Email ='use2r@gmail.com' Password='Pass123@'	Logged in successfully and redirect to user dashboard / admin dashboard	Logged in successfully and redirect to user dashboard / admin dashboard	Pass

Table 4 User Reservation Booking Page Test Case

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
1	Empty form submission	Travel Date = null Seat = '' Passenger Name = '' Contact No = ''	Warning Message popup	Warning Message popup	Pass
2	User enters invalid travel date	Travel Date = 2045-01-01 Seat = '1,2 ,3' Passenger Name = 'Sangam' Contact No = '9876543210'	Warning Message popup for date validation	Warning Message popup for date validation	Pass
3	User enters invalid phone no	Travel Date = 2026-01-01 Seat = '1,2 ,3' Passenger Name = 'Sangam' Contact No = '09998'	Error Message popup for invalid phone no	Error Message popup for invalid phone no	Pass
4	User enters valid details	Travel Date = 2026-01-01 Seat = '1,2 ,3' Passenger Name = 'Sangam' Contact No = '9876543210'	Reservation Confirmed successfully and redirect to dashboard	Reservation Confirmed successfully and redirect to dashboard	Pass

4.2.2 Test Case for System Testing

System testing evaluates the fully integrated Online Bus Reservation System (OBRS) as a whole, verifying that all modules — user registration, login, bus search, seat reservation, e-ticket generation, and admin management — function correctly and interact seamlessly. The following test cases cover each module of the system, documenting the test scenario, input data, expected output, actual output, and result for each case.

Table 5 System Testing for OBRS

ID	Test Case Description	Test Data	Expected Outcome	Actual Result	Status
TC-01	Register with all valid inputs	Name: Udesb Regmi, Email: udesb@gmail.com, Phone: 9812345678, Password: Test@1234	Account created successfully; redirected to login page	Account created; redirected to login	Pass
TC-02	Register with already existing email	Email: udesb@gmail.com (duplicate)	Error: Email already registered	Error message displayed	Pass
TC-03	Login with valid credentials	Email: sangam@gmail.com, Password: Test@1234	Login successful; redirected to User Dashboard	Dashboard loaded correctly	Pass
TC-04	Login with empty email field	Email: (blank), Password: Test@1234	Validation error: Email is required	Validation error shown	Pass
TC-05	Admin login with admin credentials	Email: admin@bus.com, Password: Admin@123	Login successful; redirected to Admin Dashboard	Admin Dashboard loaded	Pass
TC-06	User attempts to access admin panel directly via URL	URL: /admin/dashboard.php (not logged in as admin)	Access denied; redirected to login page	Redirected to login	Pass

TC-07	Search with valid source and destination	From: Kathmandu, To: Pokhara	List of available buses displayed with schedule and fare	Bus list displayed correctly	Pass
TC-08	Search with valid route but no buses available	From: Kathmandu, To: Mustang (no buses)	Message: No buses available for this route	No buses message shown	Pass
TC-09	Select an available seat	User clicks on a seat (available) seat	Seat highlighted with green as selected; proceed to confirm	Seat selected and highlighted	Pass
TC-10	Attempt to select an already booked seat	User clicks on a red (booked) seat	Seat not selectable; no action taken	Seat unclickable	Pass
TC-11	Attempt to book without selecting a seat	No seat selected; proceed clicked	Validation error: Please select a seat	Validation error shown	Pass
TC-12	View booking history after reservation	User navigates to My Reservations	All past and current bookings listed with status	Booking history displayed	Pass
TC-13	Cancel an existing booking	User selects a booking and clicks Cancel	Booking status updated to Cancelled; seat released	Cancellation processed; seat freed	Pass
TC-14	Complete payment via simulated gateway or cash	User selects Pay Now after booking confirmation	Payment processed; booking status updated to Confirmed	Status updated to Confirmed	Pass
TC-15	E-ticket generated	Completed payment for Booking ID #1045	E-ticket with passenger name, bus, seat, date, and reference	E-ticket generated correctly	Pass

			number generated		
TC-16	Download e-ticket as PDF	User clicks Download Ticket on confirmation page	PDF file downloaded with correct booking details	PDF downloaded successfully	Pass
TC-17	Print e-ticket from browser	User clicks Print Ticket	Printer dialog opens with formatted ticket layout	Print dialog opened	Pass
TC-18	E-ticket displays correct booking information	Booking: Seat 12A, Bus KTM-PKR-01, Date: 2025-11-15	Ticket shows all details: name, bus, route, seat, fare, reference	All details correctly displayed	Pass
TC-19	Booking reference number is unique per transaction	Three separate bookings made by different users	Each booking generates a unique reference number	Unique references generated	Pass
TC-20	Admin adds a new bus with valid details	Bus Number : BA 6 Pa 5919 , Bus Name: Greenline, Route From : KTM, Route To : PKR, Departure Time : 8:00AM, Arrival Time : 12:00PM, Total Seats: 32, Status : Active, Price : NPR 800, Image String: 'url/bus.jpeg'	New bus added; visible in bus list	Bus added successfully	Pass
TC-21	Admin adds bus with empty required fields	Bus Name: (blank), other fields valid	Validation error: Bus name is required	Validation error shown	Pass

TC-22	Admin edits existing bus details	Change fare of Bus ID 3 from NPR 800 to NPR 950	Bus fare updated in database and reflected on search results	Fare updated correctly	Pass
TC-23	Admin deletes a bus with no existing reservations	Delete Bus ID 7 (no bookings)	Bus removed from system; no longer appears in search	Bus deleted successfully	Pass
TC-24	Admin views complete list of all buses	Admin navigates to Manage Bus page	All buses listed with details: name, route, seats, fare, status	Full bus list displayed	Pass
TC-25	Admin views all registered users	Admin navigates to Manage Users page	Complete list of registered users with name, email, phone displayed	User list displayed correctly	Pass
TC-26	Admin views individual user details	Admin clicks on View Details of user Udesh Regmi	Full profile and booking history of selected user shown	User detail page loaded	Pass
TC-27	Admin views all reservations	Admin navigates to Manage Reservations page	All bookings listed with passenger name, bus, seat, status, and date	All reservations displayed	Pass
TC-28	Admin cancels a passenger reservation	Admin selects Booking #1032 and clicks Cancel	Booking status updated to Cancelled; seat released for rebooking	Cancellation processed	Pass
TC-29	Admin views reservation details for a specific booking	Admin clicks on Booking ID #1045	Full booking detail shown: passenger, bus, seat, fare, payment status	Booking detail displayed correctly	Pass

TC-30	Admin dashboard shows summary statistics	Admin logs in and views dashboard	Total users, total bookings, active reservations shown accurately	Dashboard statistics correct	Pass
TC-31	SQL injection attempt in login form	Email: ' OR '1'='1, Password: anything	Login rejected; no unauthorised access granted	Access denied; error shown	Pass
TC-32	Access protected page without login	Navigate to /user/dashboard.php without session	Redirected to login page	Redirected correctly	Pass
TC-33	Password stored as hash in database	New user registers with password Test@1234	Database stores hashed value, not plain text	Hash stored in users table	Pass
TC-34	System accessible on mobile browser	Open OBRS on mobile Brave browser	All pages render responsively without horizontal scroll	Responsive layout confirmed	Pass

Chapter 5: Conclusion & Future Recommendations

5.1. Lesson Learnt Outcome

The development of the Online Based Bus Reservation System (OBRS) has been a valuable learning experience throughout the BCA 4th semester project. Key lessons include mastering the integration of frontend technologies (HTML, CSS, JavaScript, Bootstrap) with backend PHP and MySQL for dynamic web applications, emphasizing the importance of secure coding practices such as prepared statements to prevent SQL injection and session management for authentication. Working within constraints—no external APIs, limited to open-source tools, and a two-member team—taught efficient time management, modular coding, and the practical application of the Waterfall methodology.

The primary outcome is a fully functional, web-based prototype tailored to Nepal's transportation needs, featuring dual client-admin modules with complete CRUD operations. The system successfully demonstrates real-time bus searching, seat booking, e-ticket generation, and administrative oversight, achieving objectives of reducing manual inefficiencies and enhancing accessibility for passengers across routes serviced by operators.

5.2 Conclusion:

In conclusion, the Online Bus Reservation System (OBRS) effectively addresses the persistent challenges in Nepal's traditional bus ticketing processes, such as long queues, overbooking, and limited real-time information, particularly in a geographically diverse nation reliant on road transport. By leveraging accessible open-source technologies—PHP for robust backend logic, MySQL for reliable data management, and responsive frontend design—the system provides a practical, cost-effective solution that supports both passengers and operators without dependency on external services.

The implementation of client features (route search, seat selection, simulated payments, and e-ticket issuance) alongside comprehensive admin tools (bus management, reservation oversight, and reporting) ensures end-to-end reservation handling. Thorough feasibility analysis, structured design (ER diagrams, DFDs up to Level 1, three-tier architecture), and rigorous testing confirmed the system's technical viability, operational usability, and performance reliability, with near-perfect functionality in simulated environments.

OBRS aligns with Nepal's growing digital infrastructure, complementing existing platforms while offering potential for small-scale operators and public entities like Sajha Yatayat. This project contributes meaningfully to the digitization of public transportation, promoting efficiency, transparency, and inclusivity, and fulfills the academic objectives of the BCA program by demonstrating applied software engineering principles in a real-world Nepali context.

5.3 Future Recommendations:

While the current prototype meets core requirements, several enhancements are recommended for future development to increase scalability, security, and user value:

1. **Integration of Live Payment Gateways:** Replace simulated payments with real integrations (e.g., eSewa, Khalti or Mobile Banking) to enable actual transactions, making the system production-ready for commercial use.
2. **Mobile Application Development:** Extend the web platform to native Android/iOS apps using frameworks like Flutter or React Native, improving accessibility for Nepal's mobile-first users.
3. **Advanced Features:**
 - Implement GPS-based bus tracking and push notifications for delays.
 - Add multilingual support (Nepali, English, and regional languages) and accessibility features for differently-abled users.
 - Incorporate basic analytics or machine learning for demand forecasting and dynamic pricing during peak seasons.
4. **Security and Performance Enhancements:** Introduce HTTPS enforcement, CAPTCHA for forms, and load balancing for high-traffic handling during festivals like Dashain.
5. **Deployment and Partnerships:** Host on reliable servers and seek collaboration with operators such as Dhading Mandali Yatayat Pvt Ltd, Sakira Travels, Jagadamba Tours and Travels or regional bus associations for real-world pilot testing and adoption.

These improvements would transform OBRS from an academic prototype into a scalable, impactful solution driving further digital transformation in Nepal's public transportation sector.

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APPENDICES

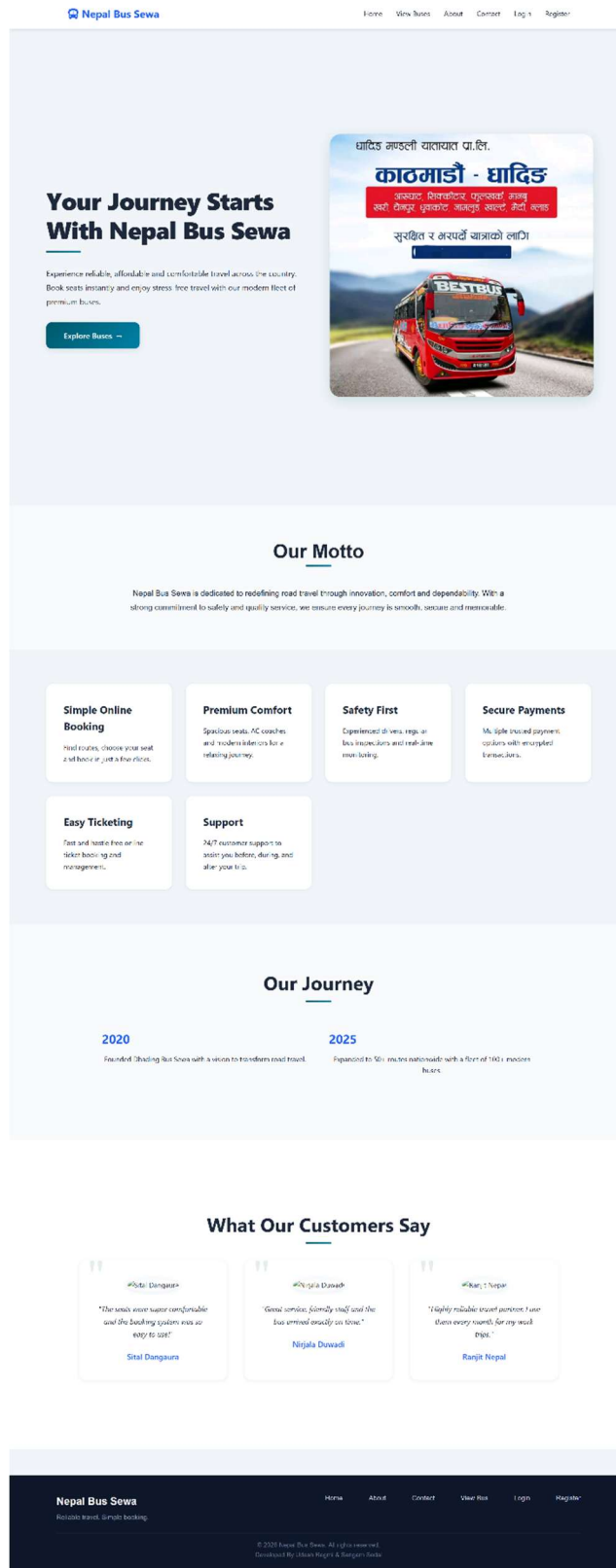


Figure 10 Home Page

Available Buses

From (e.g., Kathmandu)

To (e.g., Dhading)

Q Search

**Gangotri
AC**

Ba Pr-01-006 Kha-5217

Janakpur → Kathmandu

10:00 AM → 08:00 PM
40 seats **Rs. 1,400.00**

Login to Book

**Simrik AC**

Ba Pr-01-06 Kha-7176

Kathmandu → Kakarvitta

05:00 AM → 10:00 PM
40 seats **Rs. 2,000.00**

Login to Book

**Kamana AC**

G-AA-0061

Butwal → Kathmandu

08:00 AM → 05:00 PM
40 seats **Rs. 1,499.00**

Login to Book

**Pachpokhari Air
Bus**

Ba Pr-01-06 6188

Kakarvitta → Kathmandu

06:00 AM → 11:00 PM
40 seats **Rs. 1,999.00**

Login to Book

**King Air
Bus**

Su Pr-02-006 Kha 4444

Dang → Kathmandu

06:00 PM → 08:00 AM
40 seats **Rs. 1,699.00**

Login to Book

**Blue Best
Bus**

Ba Pr-01-006 Kha 4855

Kathmandu → Dhading

10:00 AM → 02:00 PM
40 seats **Rs. 320.00**

Login to Book

**Orange
Deluxe**

Ba Pr-02-002 Kha 1111

Dhading → Kathmandu Bus Park

06:00 AM → 09:00 AM
40 seats **Rs. 300.00**

Login to Book

**Best Bus**

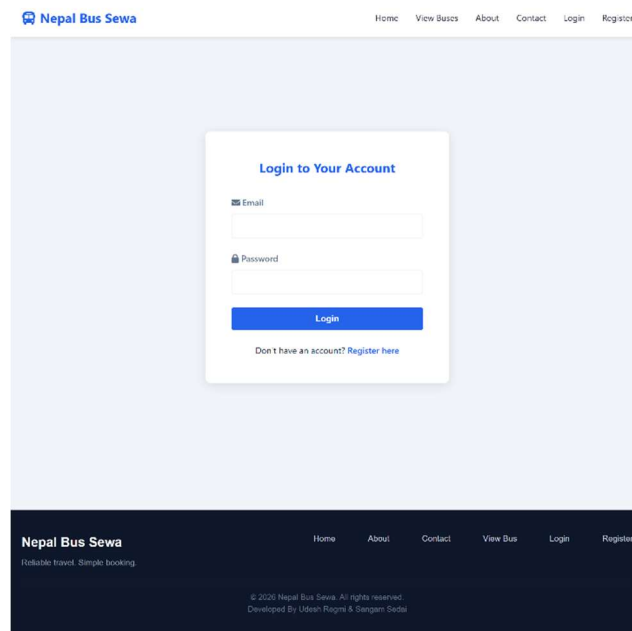
Ba Pr-02-006 5909

Kathmandu → Dhading Koirale Chautari

06:00 AM → 09:00 AM
40 seats **Rs. 400.00**

Login to Book

Figure 11. View Bus Page



The screenshot shows the login page of the Nepal Bus Sewa website. At the top, there is a navigation bar with the logo and links for Home, View Buses, About, Contact, Login, and Register. The main content area features a central white box with the title "Login to Your Account". Inside this box, there are input fields for Email and Password, each with a corresponding icon (envelope for email, lock for password). Below these fields is a blue "Login" button. At the bottom of the box, there is a link that says "Don't have an account? Register here". The footer contains the website name, tagline, navigation links, and copyright information.

Nepal Bus Sewa

Home View Buses About Contact Login Register

Login to Your Account

Email

Password

Login

Don't have an account? [Register here](#)

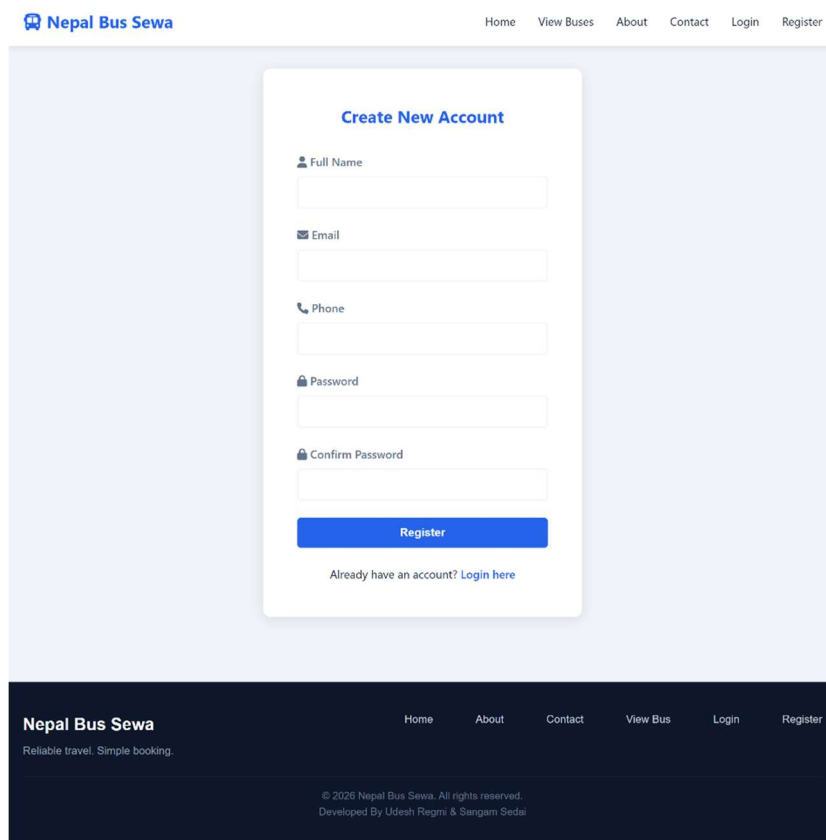
Nepal Bus Sewa

Reliable travel. Simple booking.

Home About Contact View Bus Login Register

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Developed By Udesk Regmi & Sangam Sedai

Figure 12 Login Page



The screenshot shows the register page of the Nepal Bus Sewa website. At the top, there is a navigation bar with the logo and links for Home, View Buses, About, Contact, Login, and Register. The main content area features a central white box with the title "Create New Account". Inside this box, there are input fields for Full Name, Email, Phone, Password, and Confirm Password, each with a corresponding icon (person for full name, envelope for email, phone for phone, lock for password, and lock for confirm password). Below these fields is a blue "Register" button. At the bottom of the box, there is a link that says "Already have an account? Login here". The footer contains the website name, tagline, navigation links, and copyright information.

Nepal Bus Sewa

Home View Buses About Contact Login Register

Create New Account

Full Name

Email

Phone

Password

Confirm Password

Register

Already have an account? [Login here](#)

Nepal Bus Sewa

Reliable travel. Simple booking.

Home About Contact View Bus Login Register

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Figure 13 Register Page

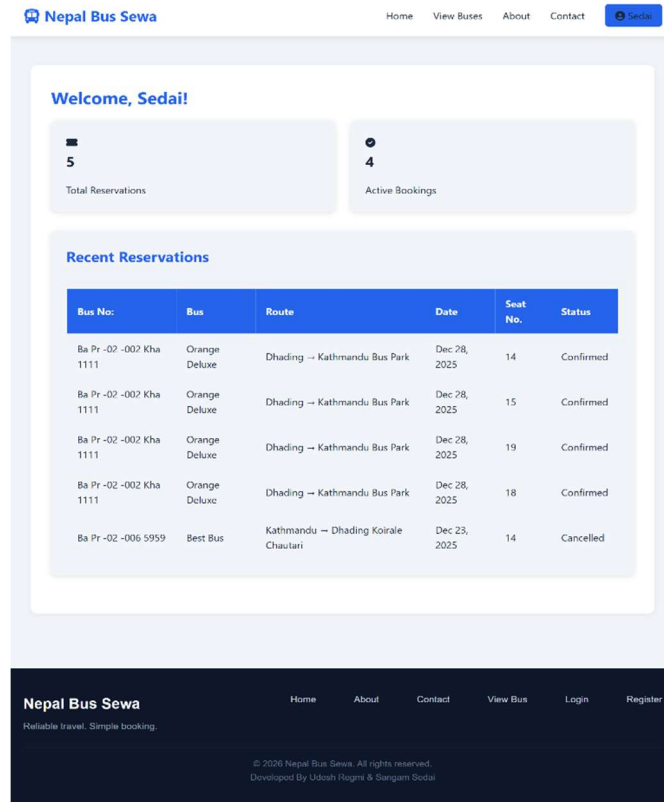


Figure 14. User Dashboard

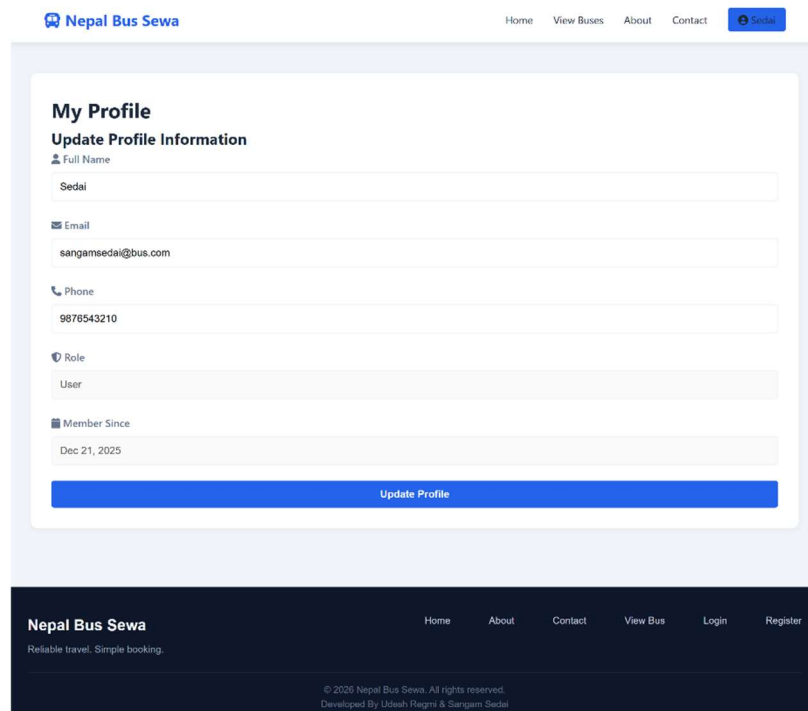


Figure 15 User Profile Page

Nepal Bus Sewa

[Home](#)
[View Buses](#)
[About](#)
[Contact](#)
[eSewa](#)

Make Reservation

Simrik AC

Ba Pr: 01-06 Kha: 7176

Kathmandu — Pokhara

05:00 AM

Rs. 2000.00 per seat

Travel Date

09/03/2026

Simrik AC Ba Pr: 01-06 Kha: 7176

40 Seats total

0 Reserved 40 Available

Choose Your Seats (Multiple Selection Allowed)

Selected seats: 8, 10, 11, 18, 23, 31

Click seats to select/deselect. You can select multiple seats.

ENTRY

DRIVER

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40

LUGGAGE COMPARTMENT

Available

Selected

Reserved

Window

Aisle

Passenger Name

Sedai

Contact Number

1234567890

Payment Method

Cash Payment

eSewa

Total Amount: Rs. 12000.00

Number of seats: 6

Confirm Reservation

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Home

About

Contact

View Bus

Login

Register

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Figure 16 Book Bus Page

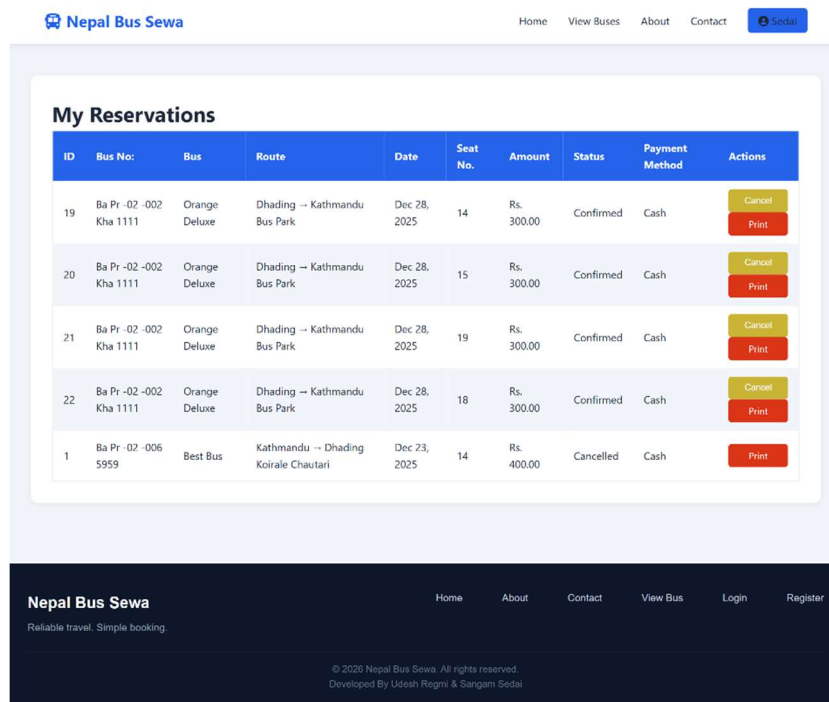


Figure 17 User Reservation Details Page

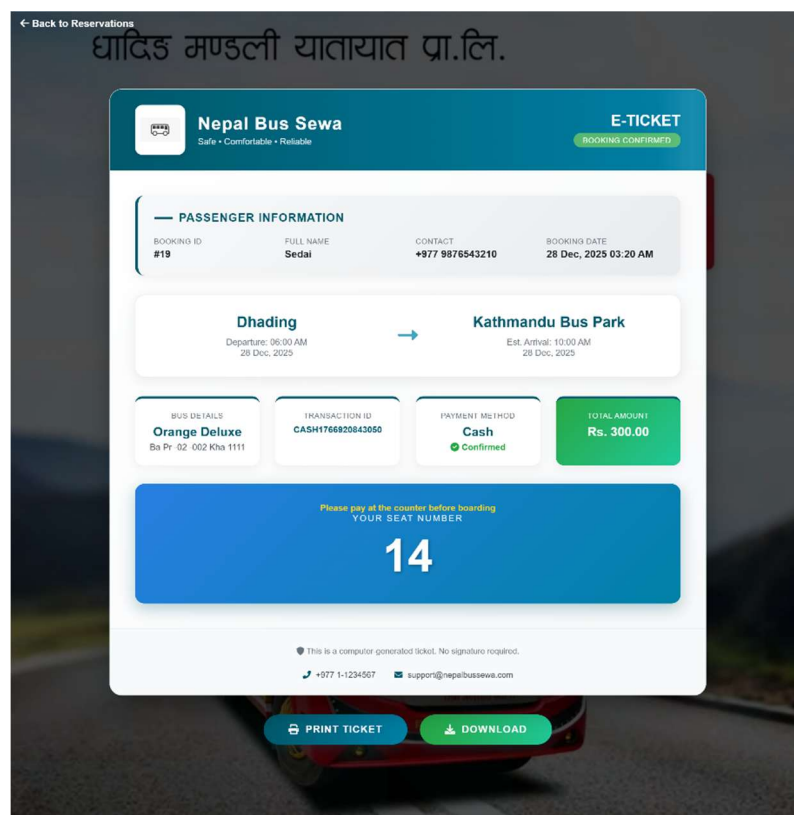


Figure 18 Print & Download Ticket Page

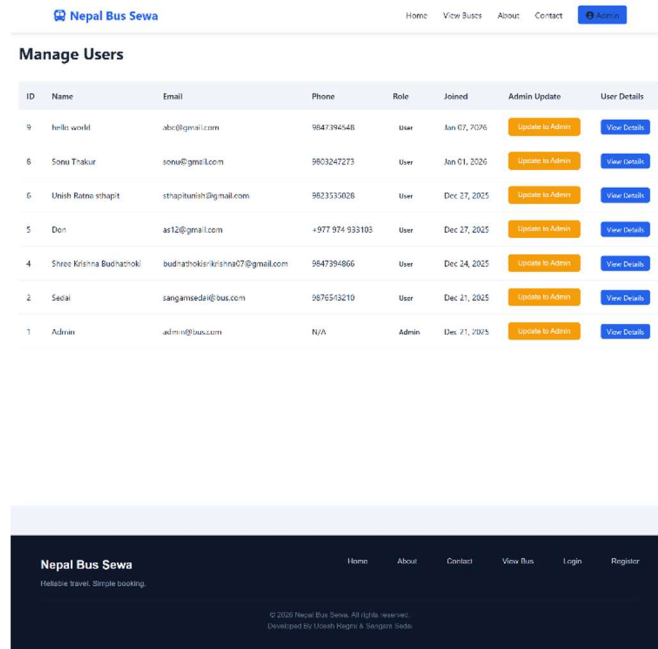


Figure 19 Admin Manage User Page

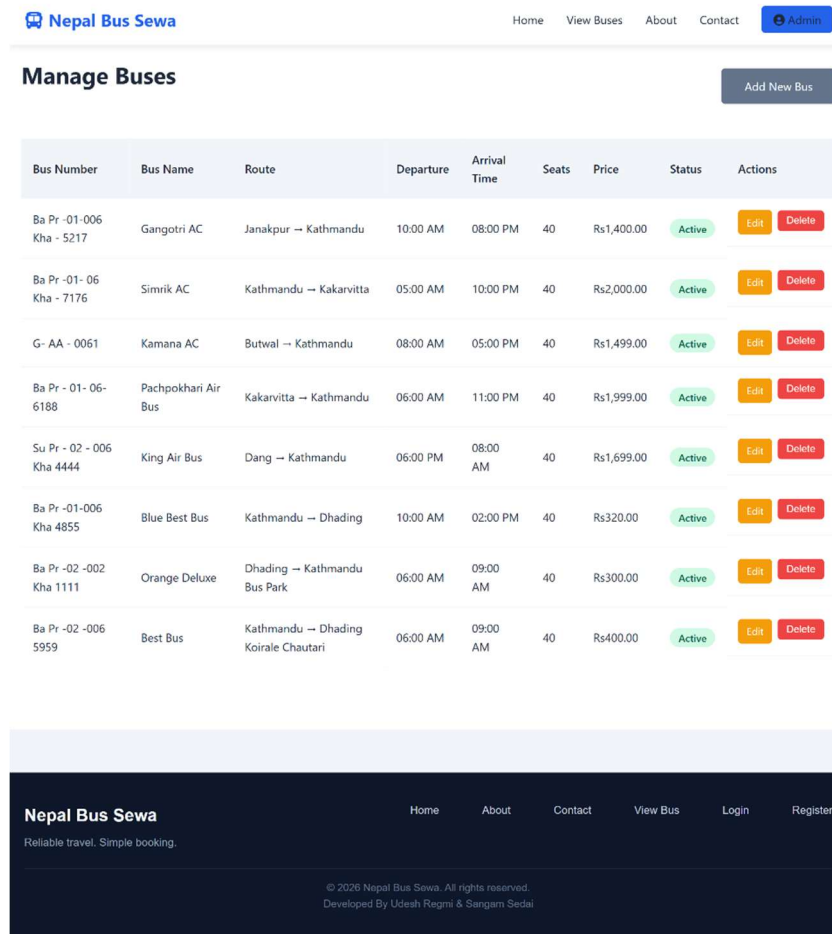


Figure 20 Admin Manage Bus Page

```

1 CREATE DATABASE IF NOT EXISTS bus_reservation CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
2 USE bus_reservation;
3
4 -- Users Table
5 CREATE TABLE users (
6     id INT AUTO INCREMENT PRIMARY KEY,
7     name VARCHAR(100) NOT NULL,
8     email VARCHAR(100) UNIQUE NOT NULL,
9     password VARCHAR(255) NOT NULL,
10    phone VARCHAR(15),
11    role ENUM('user', 'admin') DEFAULT 'user',
12    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
13    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
14    INDEX idx_email (email),
15    INDEX idx_role (role)
16 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
17
18 -- Buses Table
19 CREATE TABLE buses (
20     id INT AUTO INCREMENT PRIMARY KEY,
21     bus_number VARCHAR(50) UNIQUE NOT NULL,
22     bus_name VARCHAR(100) NOT NULL,
23     route_from VARCHAR(100) NOT NULL,
24     route_to VARCHAR(100) NOT NULL,
25     departure_time TIME NOT NULL,
26     arrival_time TIME NOT NULL,
27     total_seats INT NOT NULL,
28     available_seats INT NOT NULL,
29     price DECIMAL(10,2) NOT NULL,
30     status ENUM('active', 'inactive') DEFAULT 'active',
31     image_string VARCHAR(255) DEFAULT NULL,
32     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
33     updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
34     INDEX idx_status (status),
35     INDEX idx_route (route_from, route_to),
36     INDEX idx_bus_number (bus_number)
37 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
38
39 -- Reservations Table
40 CREATE TABLE reservations (
41     id INT AUTO INCREMENT PRIMARY KEY,
42     user_id INT NOT NULL,

```

```

-- Reservations Table
CREATE TABLE reservations (
    id INT AUTO INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    bus_id INT NOT NULL,
    seat_number INT NOT NULL,
    booking_date DATE NOT NULL,
    passenger_name VARCHAR(100) NOT NULL,
    passenger_phone VARCHAR(15) NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    status ENUM('pending', 'confirmed', 'cancelled') DEFAULT 'pending',
    payment_method ENUM('cash', 'esewa') DEFAULT 'cash',
    transaction_id VARCHAR(100) DEFAULT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    FOREIGN KEY (bus_id) REFERENCES buses(id) ON DELETE CASCADE,
    INDEX idx_user_id (user_id),
    INDEX idx_bus_id (bus_id),
    INDEX idx_booking_date (booking_date),
    INDEX idx_status (status),
    UNIQUE KEY unique_reservation (bus_id, seat_number, booking_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Insert Default Admin User
INSERT INTO users (name, email, password, role) VALUES
('Admin', 'admin@bus.com', '$2y$12$EChfFGTbi1YIWase0DibuessFKS494f.B3fY9GaR2j07wnYA3lcsa', 'admin');
-- Password: Password@123

-- Create Views for Reports
CREATE VIEW reservation_summary AS
SELECT
    DATE(r.created_at) as date,
    COUNT(*) as total_reservations,
    SUM(r.total_amount) as total_revenue,
    COUNT(DISTINCT r.user_id) as unique_customers
FROM reservations r
WHERE r.status = 'confirmed'
GROUP BY DATE(r.created_at)
ORDER BY date DESC;

```

```

1 sql
2 CREATE VIEW bus_utilization AS
3 SELECT
4     b.id,
5     b.bus_name,
6     b.bus_number,
7     b.total_seats,
8     b.available_seats,
9     (b.total_seats - b.available_seats) as occupied_seats,
10    ROUND((b.total_seats - b.available_seats) / b.total_seats * 100, 2) as occupancy_rate
11 FROM buses b
12 WHERE b.status = 'active';
13
14 -- Stored Procedure for Booking Statistics
15 DELIMITER //
16 CREATE PROCEDURE GetBookingStats(IN start_date DATE, IN end_date DATE)
17 BEGIN
18     SELECT
19         DATE(booking_date) as date,
20         COUNT(*) as bookings,
21         SUM(total_amount) as revenue,
22         AVG(total_amount) as avg_booking_value
23     FROM reservations
24     WHERE booking_date BETWEEN start_date AND end_date
25     AND status = 'confirmed'
26     GROUP BY DATE(booking_date)
27     ORDER BY date;
28 END //
29 DELIMITER ;
30
31 -- Trigger to update available seats
32 DELIMITER //
33 CREATE TRIGGER after_reservation_insert
34 AFTER INSERT ON reservations
35 FOR EACH ROW
36 BEGIN
37     IF NEW.status = 'confirmed' THEN
38         UPDATE buses
39         SET available_seats = available_seats - 1
40         WHERE id = NEW.bus_id;
41     END IF;
42 END //
43
44 CREATE TRIGGER after_reservation_cancel
45 AFTER UPDATE ON reservations
46 FOR EACH ROW
47 BEGIN
48     IF OLD.status = 'confirmed' AND NEW.status = 'cancelled' THEN
49         UPDATE buses
50         SET available_seats = available_seats + 1
51         WHERE id = NEW.bus_id;
52     END IF;
53 END //
54
55 END //

```

Figure 21 SQL Code of OBRS